

Estela Pérez Luque, PhD

Data Scientist | Simulation Validation & Human Factors

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PROFILE

I am a data scientist with a PhD in Informatics and 5+ years of experience in simulation validation and data-driven evaluation within automotive engineering. I design reproducible analysis pipelines that transform experimental and simulation data into traceable evidence for testing decisions, model benchmarking, and system improvement. My work combines predictive modelling, time-series analysis, and rigorous evaluation on real-world datasets using Python and MATLAB. I am particularly motivated to contribute to Autonomous Systems V&V at Einride, where efficient, data-driven testing is essential for building safe and reliable systems.

TECHNICAL EXPERTISE

Core Stack: Python, MATLAB; time-series analysis, predictive modelling, simulation validation, end-to-end data pipelines, model benchmarking, motion capture datasets

Verification & Validation: Scenario-based evaluation, test planning support, benchmarking protocols, validation against experimental datasets, verification metrics, traceable evidence generation

Simulation & Modelling: Simulation-based validation workflows, digital human modelling, model comparison across simulation and real-world datasets, optimization

Experimental Systems & Data Collection: Motion capture systems (MOVE4D, Xsens, optical), 4D body scanning, experimental protocol design, synchronized data collection, dataset preparation, quality control

Engineering Tools: IPS IMMA, Siemens Jack, Anybody, Autodesk Inventor, SolidWorks

Domain Strengths: Automotive ergonomics, human behavior modelling, human-product interaction, simulation credibility, cross-functional technical communication

Languages: English (C1), Spanish (Native), Swedish (B1)

PROFESSIONAL EXPERIENCE

Research Engineer (Simulation & Human Factors), University of Skövde (Sweden) **Sep 2020 – Oct 2025**

- **Verification & Validation:** I designed and executed scenario-based validation studies for predictive models by comparing simulation outputs with motion capture and in-vehicle datasets, using RMSE and DTW to quantify accuracy and behavioral variability, and producing reproducible evidence for engineering decisions.
- **Data-Driven Evaluation & Virtual Verification:** I built reproducible Python and MATLAB pipelines to benchmark model performance across participants, tasks, and conditions, while developing simulation workflows for automotive ergonomics and human motion prediction that improved simulation credibility and supported earlier, more reliable design assessments.
- **Cross-Functional Collaboration & Reporting:** I worked with industrial and academic partners, including Volvo, Scania, University of Antwerp, and Texas Tech University, to align validation scope, interpret results, and translate technical findings into clear recommendations through reports, publications, and presentations for technical and non-technical stakeholders.

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Visiting Researcher, University of Antwerp (Belgium)

Sep 2024 – Mar 2025

- **Experimental Design & Data Collection:** I designed motion-capture and 4D body-scanning protocols for human motion validation, including ethics preparation, study planning, and structured data collection procedures.
- **Data Pipelines & Verification Support:** I developed reproducible workflows for processing and organizing advanced body-scanning and motion datasets, and I standardized documentation and preparation steps that enabled multi-site collaborators to run more consistent, data-driven validation activities.

Co-founder and R&D Lead, Swe-Ex AB (Startup) (Sweden)

Sep 2021 – Mar 2023

- **Requirements Verification & Prototype Testing:** I translated user needs and safety considerations into measurable design targets and prototype test criteria, then drove three full prototype iterations from CAD design to physical validation using 3D-printed, textile, and steel components.
- **Engineering Delivery:** I combined product development, testing, and stakeholder feedback in a fast-moving startup environment, generating practical evidence to guide design decisions and improving the product through iterative verification of core concepts.

Research Assistant, University of Skövde (Sweden)

Jun 2019 – Sep 2020

- **Stakeholder Support and Delivery:** I contributed to applied ergonomics projects with 6+ industry partners by conducting analysis, producing decision-ready reports and educational documentation (including video content), and presenting findings to cross-functional teams. I helped partners apply ergonomic insights in product development discussions and translate findings into practical deliverables.

EDUCATION

PhD in Informatics, University of Skövde (Sweden)

Sep 2020 – Oct 2025

[PhD Thesis](#): Human Posture and Motion Prediction for Automotive Ergonomics Design.

Developed and validated methods to improve digital human modelling tools by benchmarking predictive models against real-world datasets using time-series metrics and simulation-based evaluation.

Master of Science in Intelligent Automation, University of Skövde (Sweden)

Aug 2018 – Jun 2019

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden)

Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain)

Aug 2012 – Jun 2018

SELECTED TECHNICAL PUBLICATION

- **Pérez Luque, E. et al. (2026).** Predicting Human Upper Extremity Reaching Motions: Comparison of Optimization-Based Method and Heuristic Method. International Journal of Human-Computer Interaction, 1–26. <https://doi.org/10.1080/10447318.2026.2632154> → Introduced a novel 3D predictive framework and multi-metric evaluation methodology for time-series model validation.

Full list available at: [Google Scholar – Estela Pérez Luque](#)